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18.100A

Real Analysis 18.100A — Real Analysis May 2026 **EXTREMELY HARD – MIT****LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Prove using the epsilon-delta definition that the function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ is differentiable at $x = 0$ but $f'(x)$ is not continuous at $x = 0$. Carefully justify all steps in your epsilon-delta arguments.
2. (3 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1 .
3. (2 points) Let M be a compact oriented n -manifold with boundary ∂M . State and prove Stokes' theorem for differential forms. Then apply it to prove the divergence theorem: $\int_M \operatorname{div} F \, dV = \int_{\partial M} F \cdot n \, dS$.
4. (4 points) Prove the Poincaré inequality: for $u \in W_0^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^n$ bounded, $\|u\|_{L^p} \leq C \|\nabla u\|_{L^p}$. Show that the constant C depends on the diameter of Ω and give the optimal constant for $p = 2$ on an interval.
5. (4 points) Prove that $\mathbb{R}P^3$ is diffeomorphic to $SO(3)$ by constructing explicit maps and verifying they are smooth inverses. Use this to compute $\pi_1(SO(3))$.
6. (3 points) Prove that symplectic integrators preserve the symplectic structure $\omega = \sum dq^i \wedge dp_i$ of Hamiltonian systems. Show that the Verlet (Störmer) method is symplectic and has energy error bounded over exponentially long times for integrable systems.
7. (6 points) Construct a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ that converges pointwise to $f(x) = 0$ but does NOT converge uniformly. Prove your construction works and explain why uniform convergence fails.
8. (3 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.
9. (4 points) Compute the Fourier transform of the distribution $p.v.(1/x)$ and prove that $\widehat{p.v.(1/x)}(\xi) = -i\pi \operatorname{sgn}(\xi)$. Use this to solve the Hilbert transform equation $H(u) = f$.

10. (3 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with contraction factor $k < 1$. Prove the Banach fixed-point theorem: T has a unique fixed point. Additionally, derive the error estimate $\|x_n - x^*\| \leq \frac{k^n}{1-k} \|x_1 - x_0\|$ where $x_{n+1} = T(x_n)$.
11. (5 points) Prove the Schauder fixed-point theorem using Schauder estimates and the Arzela-Ascoli theorem: a compact operator $T : X \rightarrow X$ on a Banach space has a fixed point if it maps a closed convex set into itself.
12. (4 points) Prove the Divergence theorem for smooth vector fields on compact manifolds with boundary and derive the Green's identities.
13. (3 points) Prove that the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$ (Plancherel's theorem) and extends uniquely from $L^1 \cap L^2$ to a unitary operator on L^2 .
14. (4 points) Prove that there is no continuous linear functional $L : \ell^\infty \rightarrow \mathbb{R}$ that extends the limit functional on c (convergent sequences) and satisfies $\|L\| = 1$. (Hint: use the Hahn-Banach theorem and consider the Banach limit.)
15. (3 points) Prove the Sobolev embedding theorem: for $p > n$, $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\alpha}(\mathbb{R}^n)$ with $\alpha = 1 - n/p$. In particular, show that if $u \in W^{1,p}(\mathbb{R}^n)$ with $p > n$, then u is (almost everywhere equal to) a Hölder continuous function.
16. (6 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω , $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.
17. (5 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \overline{B}^n \rightarrow \overline{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.
18. (5 points) Prove the open mapping theorem: if $T : X \rightarrow Y$ is a bounded linear operator between Banach spaces and T is surjective, then T maps open sets to open sets. Use it to prove the inverse mapping theorem.
19. (4 points) State and prove the spectral theorem for compact self-adjoint operators on a Hilbert space. Show that such an operator has a countable set of eigenvalues that accumulate only at zero, and the eigenvectors form an orthonormal basis.
20. (4 points) Determine all values of $p \in \mathbb{R}$ for which the series $\sum_{n=2}^{\infty} \frac{1}{n^p \ln n}$ converges. Prove your answer using the integral test and comparison tests.
21. (3 points) State and prove the Implicit Function Theorem. Apply it to show that the equation $F(x, y) = y^5 + y + x = 0$ can be solved for $y = g(x)$ near $(0, 0)$, and compute $g'(0)$.

22. (6 points) Construct an exhaustion function (proper smooth function bounded below) on any smooth manifold and use it to prove that every smooth manifold admits a complete Riemannian metric.
23. (6 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n - p)$.
24. (3 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.
25. (2 points) Prove Sard's theorem: if $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is C^k with $k > \max(n - m, 0)$, then the set of critical values of f has Lebesgue measure zero in $\mathbb{R}^m$.

Solutions

Solution 1:

We prove $f'(0) = 0$ by showing $|(f(h) - f(0))/h - 0| = |h \sin(1/h)| \leq |h|$. For any $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|h| < \delta$ implies $|h \sin(1/h)| < \varepsilon$, so $f'(0) = 0$. For $x \neq 0$, $f'(x) = 2x \sin(1/x) - \cos(1/x)$. This oscillates wildly near 0 since $\cos(1/x)$ has no limit. Hence f' is not continuous at 0.

Solution 2:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*} . Use the Frechet-Kolmogorov criterion for compactness in L^q : show $\|u_n(\cdot + h) - u_n\|_{L^q} \rightarrow 0$ uniformly as $h \rightarrow 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 3:

Stokes: $\int_M d\omega = \int_{\partial M} \omega$ for an $(n-1)$ -form ω . Proof: Use partition of unity to reduce to $\mathbb{R}^n$ case, where it follows from the fundamental theorem of calculus and Fubini. For divergence theorem, let $\omega = \sum_i (-1)^{i-1} F_i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$. Then $d\omega = (\sum_i \partial F_i / \partial x^i) dx^1 \wedge \cdots \wedge dx^n = \operatorname{div} F dV$. The boundary term gives $\int_{\partial M} F \cdot n dS$ by the Hodge star duality.

Solution 4:

For $u \in C_c^\infty(\Omega)$, write $u(x) = \int_{-\infty}^{x_1} \partial_1 u(t, x'') dt$. By Hölder, $|u(x)|^p \leq (\operatorname{diam} \Omega)^{p-1} \int |\partial_1 u|^p dt$. Integrate over Ω to get $\|u\|_p^p \leq (\operatorname{diam} \Omega)^p \|\nabla u\|_p^p$. On $(0, \pi)$, the optimal constant for $p = 2$ is $C = 1/\pi$, attained by $u(x) = \sin(x)$.

Solution 5:

Map $\mathbb{R}P^3 \rightarrow SO(3)$: represent a point in $\mathbb{R}P^3$ as a unit quaternion q modulo sign. The conjugation map $v \mapsto qvq^{-1}$ on pure quaternions gives a rotation in $SO(3)$. This is a smooth double cover $S^3 \rightarrow SO(3)$ with kernel $\{\pm 1\}$, inducing $\mathbb{R}P^3 \cong SO(3)$. Thus $\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}$.

Solution 6:

A one-step method Φ_h is symplectic if $(\Phi_h)^* \omega = \omega$. For Verlet: $q_{n+1/2} = q_n + \frac{h}{2} M^{-1} p_n$, $p_{n+1} = p_n - h \nabla V(q_{n+1/2})$, $q_{n+1} = q_{n+1/2} + \frac{h}{2} M^{-1} p_{n+1}$. Direct computation shows the Jacobian satisfies $J^T \Omega J = \Omega$. By the backward error analysis, the numerical solution is the exact flow of a nearby Hamiltonian, giving $O(e^{-c/h})$ energy conservation over exponentially long times.

Solution 7:

Let $f_n(x) = \max(0, 1 - n|x - 1/n|)$ (triangular spikes at $1/n$). For any x , for $n > 1/x$, $f_n(x) = 0$, so $f_n \rightarrow 0$ pointwise. But $\|f_n\|_\infty = 1$ for all n , so $\sup_{x \in [0,1]} |f_n(x) - 0| = 1 \not\rightarrow 0$, hence convergence is not uniform. This shows pointwise convergence does not imply uniform convergence.

Solution 8:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t} \|T(t)\| dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators

of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A .

Solution 9:

$\langle p.v.(1/x), \varphi \rangle = \langle p.v.(1/x), \widehat{\varphi} \rangle$. Write $p.v.(1/x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{x} \chi_{|x| > \varepsilon}$. Its Fourier transform: $\int_{|x| > \varepsilon} \frac{e^{-2\pi i x \xi}}{x} dx = -i\pi \operatorname{sgn}(\xi)$ as $\varepsilon \rightarrow 0$. The Hilbert transform $H(u) = \frac{1}{\pi} p.v. \int \frac{u(t)}{x-t} dt$ has symbol $\widehat{H(u)}(\xi) = -i \operatorname{sgn}(\xi) \widehat{u}(\xi)$, so $H^{-1} = -H$.

Solution 10:

Pick any $x_0 \in X$, define $x_{n+1} = T(x_n)$. Then $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$. By triangle inequality, $d(x_{n+m}, x_n) \leq \sum_{i=0}^{m-1} k^{n+i} d(x_1, x_0) \leq \frac{k^n}{1-k} d(x_1, x_0)$. So $\{x_n\}$ is Cauchy, converges to x^* . By continuity of T , $T(x^*) = x^*$. Uniqueness: if $Tx = x$, $Ty = y$, then $d(x, y) = d(Tx, Ty) \leq kd(x, y)$, so $d(x, y) = 0$. The error estimate follows from taking $m \rightarrow \infty$.

Solution 11:

Let $C \subset X$ be closed convex, $T(C) \subset C$, T compact. Approximate T by finite-rank operators T_n using Schauder basis. By Brouwer, each T_n has a fixed point x_n . By compactness of T , $\{T(x_n)\}$ has convergent subsequence $T(x_{n_k}) \rightarrow y$. Since $x_{n_k} = T_{n_k}(x_{n_k})$ and $T_{n_k} \rightarrow T$ uniformly, y is a fixed point of T .

Solution 12:

For $F \in \mathfrak{X}(\Omega)$, choose a partition of unity subordinated to a cover by coordinate charts that are half-balls near boundary. In each chart, the divergence theorem reduces to integrating a partial derivative over a cube (or half-cube), which by Fubini and FTC gives boundary integrals. Summing yields $\int_{\Omega} \operatorname{div} F dV = \int_{\partial\Omega} F \cdot n dS$. Green's identities follow by setting $F = u \nabla v$.

Solution 13:

For $f \in L^1 \cap L^2$, define $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$. Show $\int |\widehat{f}|^2 = \int |f|^2$ by proving $\int \widehat{f} \widehat{g} = \int f \bar{g}$ for g Schwartz (using Gaussian approximation and Fubini). Extend by density of $L^1 \cap L^2$ in L^2 . The inverse Fourier transform gives the adjoint, and $\mathcal{F}^* = \mathcal{F}^{-1}$.

Solution 14:

The Hahn-Banach theorem guarantees the existence of such extensions (Banach limits), but they are not unique and not continuous in the sense of being representable by a sequence. The key issue: ℓ^∞/c_0 is non-separable and the quotient map is not injective. Banach limits exist but require the axiom of choice. They satisfy $\liminf x_n \leq L(x) \leq \limsup x_n$ and are shift-invariant.

Solution 15:

For $u \in C_c^\infty(\mathbb{R}^n)$, write $u(x) - u(y)$ as an integral of ∇u over a chain connecting x and y . Using the mean value integral representation and Hölder inequality, $|u(x) - u(y)| \leq C \|\nabla u\|_{L^p} |x - y|^{1-n/p}$. By density of smooth functions in $W^{1,p}$, the inequality extends, proving the embedding is continuous.

Solution 16:

Lax-Milgram: Let $a : H \times H \rightarrow \mathbb{R}$ be coercive bilinear continuous on Hilbert H , $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$, $f(v) = \int_{\Omega} f v$. Poincaré inequality gives coercivity, trace theorem gives boundary values.

Solution 17:

Suppose f has no fixed point. Define $r(x)$ = intersection of ray from $f(x)$ through x with ∂B^n . This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y , $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y . The other endpoint would be in B^n , impossible since r maps interior to boundary. Contradiction.

Solution 18:

Use Baire category: $Y = \cup_n nT(B_X(0, 1))$ so some $nT(B_X(0, 1))$ has nonempty interior. Show $T(B_X(0, 1))$ contains a neighborhood of 0. Then by scaling, T maps open balls to open sets. The inverse mapping theorem follows: if T is bijective bounded linear, then T^{-1} is bounded.

Solution 19:

Let $T : H \rightarrow H$ be compact self-adjoint. Show $\|T\| = \sup_{|x|=1} |\langle Tx, x \rangle|$ (achieved). Let $\lambda_1 = \pm\|T\|$ be the extremal eigenvalue with eigenvector e_1 . Inductively apply the same argument to T restricted to $\{e_1, \dots, e_{n-1}\}^\perp$. Compactness implies $\lambda_n \rightarrow 0$. The eigenvectors form an ONB by the spectral decomposition $Tx = \sum \lambda_n \langle x, e_n \rangle e_n$.

Solution 20:

By the integral test, $\int_2^\infty \frac{dx}{x^p \ln x}$ converges iff $p > 1$. For $p = 1$, $\int_2^\infty \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \ln(\ln b) - \ln(\ln 2) = \infty$, so diverges. For $p < 1$, compare with $1/x^p$: $\frac{1}{n^p \ln n} > \frac{1}{n^p \cdot n^\epsilon}$ for large n , which diverges by p-test. For $p > 1$, $\frac{1}{n^p \ln n} < \frac{1}{n^{(p+1)/2}}$ for large n , which converges.

Solution 21:

IFT: Let $F : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$ be C^1 , $F(a, b) = 0$, and $\partial F / \partial y(a, b)$ invertible. Then there exists $g : U \rightarrow \mathbb{R}^n$ C^1 near a with $F(x, g(x)) = 0$ and $Dg(x) = -(\partial F / \partial y)^{-1} \partial F / \partial x$. For $F(x, y) = y^5 + y + x$, $\partial F / \partial y = 5y^4 + 1$, at $(0, 0)$ this is $1 \neq 0$, so IFT applies. $g'(0) = -(\partial F / \partial y)^{-1} \partial F / \partial x = -1/1 = -1$.

Solution 22:

Let $\{U_i\}$ be a countable cover by coordinate balls, $\{\rho_i\}$ a partition of unity. Define $f = \sum_i \rho_i$. This is proper: $f^{-1}([-\infty, c])$ is compact (only finitely many $i \leq c$ contribute). For a Riemannian metric, start with any metric g_0 , let $g = g_0 + df \otimes df$ on $f^{-1}((-\infty, \infty))$. The gradient of f has unit length, so any curve must escape to infinity in infinite length.

Solution 23:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \dots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod (\int |\partial_i u|)^{1/n} \leq \frac{1}{n} \sum \int |\partial_i u|$. For general p , apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 24:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M >$

$\max_{\partial\Omega} u^+$. At x_0 , $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$, $w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 25:

Cover U by cubes. On each cube where $\|Df\| \leq C$ and $|\det Df| \leq \varepsilon$ on a critical point, use the Whitney extension to show the image can be covered by $O((\varepsilon/C)^m)$ balls of radius $C/\sqrt{n}$. Summing over cubes gives arbitrarily small total measure. The differentiability condition ensures the critical set has enough structure.